Classification

So far we have studied regression methods, which assume the response variable is quantitative.
In many situations the response is qualitative or categorical.

Classifying: Predicting a qualitative response for an observation

Examples: some real-life examples of classification in everyday scenarios.

Our general setup will be having a training set $\{(x_1, y_1), \dots, (x_n, y_n)\}$ where x_i are our predictors (can be a mix of categorical + quantitative) and y_i is the class associated to x_i.

Our guiding example: need to create a synthetic dataset equivalent to the Default dataset where there are two categories: defaults payment or not as the categorical response variable.

K Nearest Neighbors (KNN)

Split your observations into training and test data.
Choose an integer $K \geq 1$.
Take one of your observations x_i in the test set.
The KNN classifier does the following to assign a class to x_i:

1. identify the K points in the training data closest to x_i
2. identify the majority class amongst those K nearest neighbors
3. assign this majority class to x_i

Add an example like ISLP figure 2.14 where students can calculate themselves a couple of classes for KNN with K=3.

Pros / cons of using KNN for classification

Accuracy assessment for classifiers

(compare against null classifier)

Take an observation (x_0, y_0) in the test set.

We have the following possibilities depending on the class y_0 belongs to and what prediction was given based on the predictors x_0.

	x_0 got classified as POSITIVE class	x_0 got classified as NEGATIVE class
y_0 is POSITIVE class	True positive	False negative
y_0 is NEGATIVE class	False positive	True negative

So each of our observations in the test set can be classified as true / false positive or true / false negative.

Check-in: where students categorize a few made-up classifications as true / false positives or true / false negatives.

Looking at all the classification results on our test data, we can count the total number of true / false positives and true / false negatives:

TP = total # of true positives
TN = " " " true negatives
FP = " " " false positives
FN = " " " false negatives

This information can be organized into a confusion matrix:

true class \ predicted class	positive	negative
positive	TP	FN
negative	FP	TN

Some popular accuracy metrics we can calculate given this information are:

overall accuracy: $\dfrac{TP + TN}{TP + FP + TN + FN}$: fraction of correctly classified samples

precision: $\dfrac{TP}{TP + FP}$: out of what was classified as P, what fraction is really P?

recall: $\dfrac{TP}{TP + FN}$: what fraction of P was correctly classified?

Note: add alternative names for precision and recall.

Sometimes it can be convenient to combine precision and recall into a single metric called the F_1 score.

$$F_1 = 2\,\frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

- Simple way to compare different classifiers
- Favors classifiers with similar precision and recall: you may not always want this!
- High F_1 score happens when you have both high precision and recall.

Check-in: I need exercises here for students to interpret and compare confusion matrices, precision, recall, and F_1 score.
Probably comparing KNN to the null classifier.

Explain how these errors can also be computed on the training set and how test error always tends to be larger than training error (as before).

Overfitting

Remember: a model is overfitting the data when it has a small training error, but a large test error.

Check-in: training and test error for KNN when K=1.

Similarly to the regression setting, when we use more flexible classification methods the train error may go down, but the test error may not.

[Show three plots with different KNN parameters: K=1, 20, 100. Have this be similar to Figure 2.16]

Check-in: have students identify overfitting from the graphs

Show the train-test MSE plot same as in previous lesson. Similar to figure 2.17.
End with bias-variance discussion.

Next lesson:

- Logistic regression, multiple logistic regression.
- Precision, recall tradeoff
- Receiver operator curve
- AUC interpretation
- Linear discriminant analysis (?)
- Comparing models.

* will use same example dataset.

Comparing KNN, logistic and LDA. ISLP page 164.